
CURRENT ONE-STAGE AND TWO-STAGE IPDE DESIGN WITH EFFICACY

Updated carryover-model options and time-to-event implementation

Purpose. This document updates the efficacy-enabled AIDE/IPDE framework to include two toxicity carryover models, four efficacy carryover models, and a time-to-event implementation that combines a fixed- r TITE-CRM likelihood for toxicity with a delayed-outcome likelihood approximation for efficacy.

Design family	AIDE/IPDE with binary toxicity and binary efficacy
Dose levels	$j = 1, \dots, J$
Trial structures	One-stage and two-stage
Toxicity options	Shared r_T complement-discount model; shared α_T additive previous-dose model
Efficacy options	r_E or α_E , each specified either dose-specifically or shared across dose levels
TITE implementation	Toxicity: fixed- r_T TITE-CRM; efficacy: delayed-outcome approximation
Enrollment schemes	Continuous enrollment with new-patient priority, or IPDE-first recruitment
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Technical design specification. The delayed-outcome efficacy component is an endpoint adaptation of the likelihood approximation developed for delayed binary outcomes; it must be evaluated through simulation for the intended efficacy assessment window and response-time distribution.

Design Summary

Core principle. Toxicity determines safety, admissibility, and the allowable direction of dose movement. Efficacy or utility ranks doses only after toxicity restrictions have been applied. An IPDE treatment is allowed only when its individual toxicity-risk criterion and expected efficacy-improvement criterion are both satisfied.

The trial structure, enrollment mechanism, carryover model, and delayed-outcome implementation are separate design choices except that the current TITE toxicity implementation is explicitly restricted to the fixed- r_T model.

Model option matrix

Endpoint	Option	Parameter scope	IPDE probability
Toxicity	r_T complement-discount	Shared across levels	$q_{T,i} = r_T + (1 - r_T)p_{T,j_i}$
Toxicity	α_T additive previous-dose	Shared across levels	$q_{T,i} = \min\{1, p_{T,j_i} + \alpha_T p_{T,\ell_i}\}$
Efficacy	$r_{E,j}$ complement-discount	Dose-specific	$q_{E,i} = r_{E,j_i} + (1 - r_{E,j_i})p_{E,j_i}$
Efficacy	$\alpha_{E,j}$ additive previous-dose	Dose-specific	$q_{E,i} = \min\{1, p_{E,j_i} + \alpha_{E,j_i} p_{E,\ell_i}\}$
Efficacy	r_E complement-discount	Shared across levels	$q_{E,i} = r_E + (1 - r_E)p_{E,j_i}$
Efficacy	α_E additive previous-dose	Shared across levels	$q_{E,i} = \min\{1, p_{E,j_i} + \alpha_E p_{E,\ell_i}\}$

Fixed versus estimated carryover. A shared or dose-specific carryover coefficient may be fixed at a prespecified value or estimated with a prespecified Beta prior. The model definitions below show the estimated version unless explicitly labeled fixed. In the current TITE toxicity implementation, $r_T = r_T^{\text{fix}}$ is supplied as data and is not sampled.

Notation

Symbol	Definition	Symbol	Definition
i	Administration-level index	j	Generic dose-level index
j_i	Current dose for administration i	ℓ_i	Same patient's immediately preceding dose
I_i	1 for IPDE/recycled administration; 0 for regular	$p_{T,j}$	Regular-patient toxicity probability at dose j
$p_{E,j}$	Regular-patient efficacy probability at dose j	$q_{T,i}$	Model-specific IPDE toxicity probability
$q_{E,i}$	Model-specific IPDE efficacy probability	ϕ_T	Target toxicity probability
ϕ_E	Minimum acceptable efficacy for futility	c_T, c_E	Posterior monitoring cutoffs
δ	Minimum expected efficacy improvement for IPDE	m_U	One-stage sample threshold
$u_{T,i}, u_{E,i}$	Follow-up accumulated by interim time	τ_T, τ_E	Toxicity and efficacy assessment windows
$w_{T,i}, w_{E,i}$	Endpoint-specific follow-up weights	$\Delta_{E,i}$	Efficacy ascertainment indicator

1. Statistical Model

1.1 Baseline toxicity CRM

Let p_{0j} be the prespecified CRM skeleton probability at dose j , and let β be the one-dimensional CRM parameter. Under a power CRM,

$$p_{T,j}(\beta) = p_{0j}^{\exp(\beta)}, \quad \beta \sim N(\mu_\beta, \sigma_\beta^2). \quad (1)$$

For a regular administration at dose j_i ,

$$Y_{T,i} \mid \beta, I_i = 0 \sim \text{Bernoulli}\{p_{T,j_i}(\beta)\}. \quad (2)$$

The IPDE probability is determined by one of the following two toxicity options.

1.2 Toxicity option T-R: shared complement-discount parameter r_T

For an IPDE administration at current dose j_i ,

$$q_{T,i}^{(r)} = r_T + (1 - r_T)p_{T,j_i}(\beta), \quad 0 \leq r_T \leq 1. \quad (3)$$

When estimated, $r_T \sim \text{Beta}(a_{r_T}, b_{r_T})$. When fixed, $r_T = r_T^{\text{fix}}$ is user-specified. The parameter is shared across dose levels, so all IPDE toxicity observations inform the same coefficient. When $r_T = 0$, the IPDE and regular probabilities are identical; increasing r_T shifts the IPDE risk toward one.

1.3 Toxicity option T-A: shared additive previous-dose parameter α_T

For an IPDE administration at current dose j_i after a preceding dose ℓ_i ,

$$q_{T,i}^{(\alpha)} = \min\{1, p_{T,j_i}(\beta) + \alpha_T p_{T,\ell_i}(\beta)\}, \quad \alpha_T \sim \text{Beta}(a_{\alpha_T}, b_{\alpha_T}). \quad (4)$$

The coefficient α_T is shared across IPDE administrations, while the added risk depends on the patient's actual preceding dose. The cap at one is applied administration by administration.

Current TITE restriction. Both T-R and T-A are complete-outcome toxicity model options. The current time-to-event specification in Section 2 uses T-R with a fixed r_T only. A TITE implementation of T-A would require a separate software extension and is not part of the present design specification.

1.4 Baseline efficacy model

The regular efficacy probability is modeled separately at each dose:

$$p_{E,j} \sim \text{Beta}(a_{E,j}, b_{E,j}), \quad j = 1, \dots, J. \quad (5)$$

Equivalently, the individual-level model is

$$Y_{E,i} \mid p_{E,j_i}, I_i = 0 \sim \text{Bernoulli}(p_{E,j_i}). \quad (6)$$

After aggregation of regular administrations at dose j , this produces the usual beta-binomial updating. Individual rows are retained in implementation because the additive IPDE options depend on the preceding dose.

1.5 Four efficacy carryover options

1.5.1 E-RD: r model with dose-specific carryover

$$r_{E,j} \sim \text{Beta}(a_{r,j}, b_{r,j}), \quad (7)$$

$$q_{E,i}^{(r,\text{dose})} = r_{E,j_i} + (1 - r_{E,j_i})p_{E,j_i}. \quad (8)$$

Each destination dose has its own $r_{E,j}$. IPDE data at dose j directly inform that dose's carryover coefficient.

1.5.2 E-AD: α model with dose-specific carryover

$$\alpha_{E,j} \sim \text{Beta}(a_{\alpha,j}, b_{\alpha,j}), \quad (9)$$

$$q_{E,i}^{(\alpha, \text{dose})} = \min \{1, p_{E,j_i} + \alpha_{E,j_i} p_{E,\ell_i}\}. \quad (10)$$

The additive coefficient is indexed by the destination dose j_i , while the amount added depends on the preceding-dose efficacy probability p_{E,ℓ_i} .

1.5.3 E-RS: r model shared across dose levels

$$r_E \sim \text{Beta}(a_r, b_r), \quad (11)$$

$$q_{E,i}^{(r, \text{shared})} = r_E + (1 - r_E) p_{E,j_i}. \quad (12)$$

One shared r_E pools IPDE information across all destination doses.

1.5.4 E-AS: α model shared across dose levels

$$\alpha_E \sim \text{Beta}(a_\alpha, b_\alpha), \quad (13)$$

$$q_{E,i}^{(\alpha, \text{shared})} = \min \{1, p_{E,j_i} + \alpha_E p_{E,\ell_i}\}. \quad (14)$$

This is the shared-parameter additive previous-dose model. Two IPDE administrations at the same destination can have different probabilities if their preceding doses differ.

1.6 Gamma-ratio representation of Beta priors

For JAGS implementations in which the direct Beta density is to be represented through Gamma variables, any parameter $c \sim \text{Beta}(a, b)$ may be written as

$$g_1 \sim \text{Gamma}(a, 1), \quad g_2 \sim \text{Gamma}(b, 1), \quad c = \frac{g_1}{g_1 + g_2}. \quad (15)$$

The Gamma variables must use a common rate. This representation applies to $p_{E,j}$ and to every shared or dose-specific r or α coefficient. It is mathematically equivalent to the intended Beta prior.

1.7 Complete-outcome likelihood

Let

$$\theta_{T,i} = (1 - I_i) p_{T,j_i} + I_i q_{T,i}, \quad \theta_{E,i} = (1 - I_i) p_{E,j_i} + I_i q_{E,i}, \quad (16)$$

where $q_{T,i}$ and $q_{E,i}$ are selected from the active model options. With complete binary outcomes,

$$L_T \propto \prod_{i=1}^{N_T} \theta_{T,i}^{Y_{T,i}} (1 - \theta_{T,i})^{1-Y_{T,i}}, \quad (17)$$

$$L_E \propto \prod_{i=1}^{N_E} \theta_{E,i}^{Y_{E,i}} (1 - \theta_{E,i})^{1-Y_{E,i}}. \quad (18)$$

For complement-discount models, rows can be grouped by current dose and administration type. For additive previous-dose models, the individual-level form is required because the pair (ℓ_i, j_i) changes the IPDE probability.

Decision summary convention. Posterior means are the default point estimates in the operating rules. Posterior medians may be substituted, but the same convention must be used across all doses, endpoints, and decision times.

2. Time-to-Event and Delayed-Outcome Implementation

2.1 Interim endpoint notation

At an interim decision time, each administration has an assigned dose, administration type, observed event indicator by that time, and accumulated follow-up. Endpoint-specific weights are clipped to $[0, 1]$. Under the default uniform event-time assumption,

$$w_{T,i} = \min \left\{ 1, \max \left(0, \frac{u_{T,i}}{\tau_T} \right) \right\}, \quad w_{E,i} = \min \left\{ 1, \max \left(0, \frac{u_{E,i}}{\tau_E} \right) \right\}. \quad (19)$$

Observed toxicity events, observed efficacy responses, and fully completed assessment windows are treated as ascertained outcomes. The toxicity and efficacy endpoints use different likelihood conventions below.

2.2 Toxicity: fixed- r_T TITE-CRM likelihood

The current TITE toxicity model uses the shared complement-discount model with a user-specified constant r_T^{fix} . Define the pre-TITE row probability

$$\theta_{T,i} = (1 - I_i)p_{T,j_i}(\beta) + I_i [r_T^{\text{fix}} + (1 - r_T^{\text{fix}})p_{T,j_i}(\beta)]. \quad (20)$$

For a DLT, $w_{T,i}$ is forced to one. The TITE-weighted event probability is

$$\theta_{T,i}^{\text{eff}} = w_{T,i}\theta_{T,i}, \quad (21)$$

and the interim likelihood is

$$L_T^{\text{TITE}}(\beta \mid r_T^{\text{fix}}) = \prod_{i=1}^{N_T} (w_{T,i}\theta_{T,i})^{Y_{T,i}} (1 - w_{T,i}\theta_{T,i})^{1-Y_{T,i}}. \quad (22)$$

Toxicity status	Convention	Likelihood contribution
Observed DLT	$Y_{T,i} = 1, w_{T,i} = 1$	$\theta_{T,i}$
Completed non-DLT	$Y_{T,i} = 0, w_{T,i} = 1$	$1 - \theta_{T,i}$
Pending non-DLT	$Y_{T,i} = 0, 0 < w_{T,i} < 1$	$1 - w_{T,i}\theta_{T,i}$
No follow-up	$Y_{T,i} = 0, w_{T,i} = 0$	1

Toxicity convention. The pending non-DLT contribution is $1 - w_{T,i}\theta_{T,i}$. The delayed-outcome approximation $(1 - \theta_{T,i})^{w_{T,i}}$ is not used for toxicity in this design. When $r_T^{\text{fix}} = 0$, the model reduces to the original TITE-CRM likelihood for regular and IPDE rows.

2.3 Efficacy: delayed-outcome likelihood approximation

Let $\theta_{E,i}$ be the row-specific efficacy probability generated by one of E-RD, E-AD, E-RS, or E-AS. Let $\Delta_{E,i} = 1$ when the efficacy endpoint is ascertained by the interim time, meaning that a response has occurred or the full efficacy window is complete; otherwise $\Delta_{E,i} = 0$ and the patient is pending without an observed response.

For a pending nonresponse, the exact event-time observed-data contribution under weight $w_{E,i} = \Pr(T_{E,i} \leq u_{E,i} \mid Y_{E,i} = 1)$ is $1 - w_{E,i}\theta_{E,i}$. The delayed-outcome approach replaces that term by the first-order approximation

$$1 - w_{E,i}\theta_{E,i} \approx (1 - \theta_{E,i})^{w_{E,i}}. \quad (23)$$

The resulting efficacy likelihood is

$$L_E^{\text{DO}} \propto \prod_{i=1}^{N_E} \theta_{E,i}^{\Delta_{E,i} Y_{E,i}} (1 - \theta_{E,i})^{\Delta_{E,i}(1-Y_{E,i}) + (1-\Delta_{E,i})w_{E,i}}. \quad (24)$$

Thus, an observed response contributes $\theta_{E,i}$, a completed nonresponse contributes $1 - \theta_{E,i}$, and a pending nonresponse contributes $(1 - \theta_{E,i})^{w_{E,i}}$.

Efficacy status	Convention	Likelihood contribution
Observed response	$Y_{E,i} = 1, \Delta_{E,i} = 1$	$\theta_{E,i}$
Completed nonresponse	$Y_{E,i} = 0, \Delta_{E,i} = 1$	$1 - \theta_{E,i}$
Pending nonresponse	$Y_{E,i} = 0, \Delta_{E,i} = 0$	$(1 - \theta_{E,i})^{w_{E,i}}$
No efficacy follow-up	$w_{E,i} = 0$	1

Endpoint adaptation and validation. The cited delayed-outcome derivation was developed and evaluated for delayed toxicity. The present design applies the same binary-event approximation to efficacy response. This extension should be validated by simulation under plausible efficacy response-time distributions, assessment-window lengths, accrual rates, and model misspecification.

2.4 Combined interim analysis

At each model update:

1. Fit the fixed- r_T TITE-CRM toxicity model using all started toxicity rows and their $w_{T,i}$ values.
2. Fit the selected efficacy carryover model using ascertained efficacy outcomes and weighted pending nonresponses through L_E^{DO} .
3. Use posterior summaries of $p_{T,j}$ for dose-level safety and MTD decisions, and model-specific $q_{T,i}$ for individual IPDE safety.
4. Use posterior summaries of $p_{E,j}$ for efficacy ranking, futility, utilities, and final OBD selection, and model-specific $q_{E,i}$ for IPDE improvement.

If the one-stage threshold is defined through partial information, the toxicity effective sample size at dose d may be taken as $n_{T,d}^{\text{eff}} = \sum_{i:j_i=d} w_{T,i}$. The software must state whether m_U is compared with this quantity or with the count of fully evaluable administrations.

3. IPDE Trial Design

At every decision time, safety is evaluated before efficacy or utility. Doses eliminated for overtotoxicity or futility are removed from cohort-allocation and IPDE-destination sets.

3.1 Two-stage design

1. **Stage I: toxicity-driven AIDE.** Treat patients according to the toxicity-based AIDE design until the cumulative Stage I sample size has reached the user-specified limit `s1_Max` and the next toxicity recommendation is *stay*. Efficacy outcomes are collected and the delayed-outcome efficacy model is updated, but Stage I cohort assignments are determined by toxicity.
2. **Continuous monitoring during Stage I.** Apply overtotoxicity, futility, and IPDE rules whenever posterior distributions are updated. An efficacy-futile dose may therefore be eliminated during Stage I even though efficacy does not drive Stage I assignment.
3. **Stage II: efficacy-driven allocation.** Among currently admissible doses, assign the next cohort to the dose with the largest posterior regular-patient efficacy estimate,

$$\hat{p}_{E,j} = E(p_{E,j} \mid \mathcal{D}). \quad (25)$$

A dose eliminated for overtotoxicity or futility cannot be selected.

4. **Stage II recycling.** Evaluate each potentially recyclable patient using the destination-specific criteria in Section 4.3. When `flexible = FALSE`, recycling is upward only relative to the patient's preceding dose. When `flexible = TRUE`, any admissible destination may be considered.

Stage II ranking set. Efficacy comparison is made only among admissible doses. Tried-dose restrictions, adjacency restrictions, dose-skipping restrictions, and any MTD ceiling are applied before identifying the dose with the highest estimated efficacy.

3.2 One-stage design

Let d be the current trial dose and n_d the active toxicity sample quantity at d (fully evaluable count or prespecified effective sample size). Let m_U be the user-specified threshold; the current working example is $m_U = 6$.

The toxicity recommendation is obtained first.

Toxicity decision	Condition	Cohort-allocation rule
De-escalate	Any n_d	De-escalate according to the toxicity rule. Utility cannot override direct safety-driven de-escalation.
Stay	$n_d \geq m_U$	Rank admissible doses in $\{d-1, d\}$ by utility and assign the cohort to the largest-utility dose.
Escalate or stay	Escalate, or stay with $n_d < m_U$	Rank admissible doses in $\{d-1, d, d+1\}$ by utility and assign the cohort to the largest-utility dose.

Before utility ranking, truncate candidate indices to $\{1, \dots, J\}$ and remove overtotoxic, futile, untried when required, or otherwise prohibited doses. If one candidate remains, assign that dose without a utility comparison.

3.3 Enrollment schemes

Scheme	Queue priority	Operational rule
Continuous enrollment	New patients first	Enrollment remains open. When a cohort is formed, newly enrolled patients are placed first and eligible IPDE patients fill remaining positions. Arrivals while a cohort is closed wait in the appropriate queue.
IPDE-first recruitment	Eligible IPDE patients first	When a cohort is ready, place eligible IPDE patients first. Recruit only enough new patients to complete the cohort. New-patient enrollment is activated as needed.

Event-clock operation. New-patient arrivals and completed-cycle IPDE decisions are assignable events. If no cohort position is open, the corresponding opportunity is placed in the waiting queue. The selected enrollment scheme determines queue priority; the one-stage or two-stage design determines the assigned dose.

4. Overtotoxicity, Futility, and IPDE Criteria

The following checks are repeated whenever outcomes or follow-up information change the posterior distributions.

4.1 Overtotoxicity monitoring

For each dose j , eliminate the dose for overtotoxicity when

$$\Pr(p_{T,j} > \phi_T \mid \mathcal{D}) > c_T. \quad (26)$$

An eliminated dose cannot receive a new cohort or serve as an IPDE destination. Under toxicity monotonicity, elimination of dose j may also eliminate all higher doses. The trial terminates early when no admissible safe dose remains, including when the lowest dose is eliminated.

4.2 Efficacy futility monitoring

Eliminate dose j for futility when

$$\Pr(p_{E,j} < \phi_E \mid \mathcal{D}) > c_E. \quad (27)$$

A futile dose cannot receive a new cohort, serve as an IPDE destination, or enter final OBD selection. In the TITE implementation, this posterior is calculated from the selected efficacy model using the delayed-outcome likelihood.

4.3 IPDE eligibility and destination criteria

Suppose a patient was previously treated at dose ℓ and is considered for destination dose k . Recycling is permitted only when all conditions below are satisfied.

1. Individual toxicity risk.

$$\Pr\{q_T(\ell \rightarrow k) > \phi_T \mid \mathcal{D}\} < c_T. \quad (28)$$

For the two complete-outcome toxicity options,

$$q_T^{(r)}(\ell \rightarrow k) = r_T + (1 - r_T)p_{T,k}, \quad (29)$$

$$q_T^{(\alpha)}(\ell \rightarrow k) = \min\{1, p_{T,k} + \alpha_T p_{T,\ell}\}. \quad (30)$$

Under the current TITE implementation, use $q_T^{(r)}$ with $r_T = r_T^{\text{fix}}$.

2. Expected efficacy improvement.

$$\mathbb{E}\{q_E(\ell \rightarrow k) - p_{E,\ell} \mid \mathcal{D}\} > \delta, \quad (31)$$

where q_E is computed under E-RD, E-AD, E-RS, or E-AS. The preferred implementation evaluates the posterior expectation draw by draw. A plug-in approximation may use posterior means of all model components.

3. Cycle limit.

The patient satisfies $n_{\text{cycle}} < \text{max_cycle}$.

4. Completed prior-cycle outcomes.

The cycle used to establish recycling eligibility is complete, and the patient experienced neither a DLT nor an efficacy response in that completed cycle.

5. Dose-direction restriction.

If `flexible` = FALSE, require an upward destination $k > \ell$ subject to the active dose-skipping and admissibility rules. If `flexible` = TRUE, any admissible destination may be considered.

If no destination satisfies every criterion, the patient is not recycled.

4.4 Order of operations

1. Update the toxicity posterior and the selected efficacy posterior using all currently available complete and partial information.

2. Apply dose-level overtoxicity and futility rules and update the admissible dose set.
3. Determine the next cohort dose using the active one-stage or two-stage allocation rule.
4. Evaluate each potential IPDE patient against patient- and destination-specific criteria before cohort placement.

5. Utility Functions

Utilities are based on regular-patient probabilities $p_{E,j}$ and $p_{T,j}$. Carryover parameters affect their posterior estimation through IPDE likelihood contributions, but patient-specific q_E and q_T are reserved for recycling decisions.

5.1 Utility 1: efficacy only

$$U_1(j) = p_{E,j}. \quad (32)$$

Toxicity affects the candidate set through safety rules and the MTD ceiling rather than through the numerical utility.

5.2 Utility 2: efficacy minus toxicity penalty

$$U_2(j) = p_{E,j} - \lambda_T p_{T,j}, \quad \lambda_T \geq 0. \quad (33)$$

5.3 Utility 3: four-outcome joint score

Assign score u_{te} to joint outcome $(T = t, E = e)$. Under the working-independence approximation,

$$U_3(j) = u_{00}(1 - p_{T,j})(1 - p_{E,j}) + u_{01}(1 - p_{T,j})p_{E,j} \quad (34)$$

$$+ u_{10}p_{T,j}(1 - p_{E,j}) + u_{11}p_{T,j}p_{E,j}. \quad (35)$$

Only admissible doses are ranked using a common posterior-summary convention. A prespecified tie-breaking rule is required; selecting the lower dose is the conservative default. Utility cannot override direct toxicity-driven de-escalation.

6. Final OBD Selection

Final selection is sequential: the toxicity model determines the MTD ceiling, the efficacy model removes futile doses, and utility identifies the optimal remaining dose.

1. **Select the MTD from the toxicity model.** Apply the implemented CRM MTD rule using posterior estimates of $p_{T,j}$. By default, restrict selection to tried doses and exclude doses eliminated for overtotoxicity. A standard rule selects the tried admissible dose whose posterior toxicity estimate is closest to ϕ_T .
2. **Remove efficacy-futile doses.** Exclude each dose satisfying $\Pr(p_{E,j} < \phi_E \mid \mathcal{D}) > c_E$.
3. **Construct the OBD candidate set.**

$$\mathcal{C}_{\text{OBD}} = \left\{ j : j \text{ tried, } j \leq \hat{j}_{\text{MTD}}, j \text{ neither overtotoxic nor futile} \right\}. \quad (36)$$

4. **Select the highest-utility dose.**

$$\hat{j}_{\text{OBD}} = \arg \max_{j \in \mathcal{C}_{\text{OBD}}} U(j). \quad (37)$$

If $\mathcal{C}_{\text{OBD}}$ is empty, no OBD is selected. The design must not force a selection when all tried doses are inadmissible.

Default restrictions and reporting

- Restrict MTD and OBD selection to tried doses by default.
- State any tried-dose, adjacency, or dose-skipping restrictions used in Stage II or one-stage ranking.
- Report the selected toxicity and efficacy model options, whether carryover coefficients were fixed or estimated, and all prior specifications.
- Report MTD and OBD selection, posterior dose-level estimates, utility, eliminations, IPDE counts and destinations, trial duration, regular-patient enrollment, pending-outcome handling, and no-selection outcomes.

7. Compact Algorithms and Control Parameters

Two-stage algorithm

1. Initialize the fixed- r_T TITE toxicity model and the selected delayed-outcome efficacy model at the starting dose.
2. During Stage I, allocate cohorts by toxicity-based AIDE until **s1_Max** is reached and the next toxicity recommendation is stay.
3. At every update, apply overtotoxicity, futility, and IPDE checks using current partial follow-up.
4. During Stage II, assign cohorts to the admissible dose with the largest posterior $p_{E,j}$ estimate.
5. At final analysis, select the MTD, remove futile doses, and choose the highest-utility tried dose not exceeding the MTD.

One-stage algorithm

1. Update the fixed- r_T TITE-CRM at current dose d and obtain the toxicity recommendation.
2. If de-escalation is recommended, de-escalate directly.
3. If stay is recommended and $n_d \geq m_U$, rank $\{d-1, d\}$ by utility after exclusions.
4. If escalation is recommended, or stay with $n_d < m_U$, rank $\{d-1, d, d+1\}$ by utility after exclusions.
5. Evaluate potential IPDE patients separately using model-specific destination probabilities before cohort placement.

User-specified controls

Parameter	Role	Convention
<code>tox_model</code>	Complete-outcome toxicity model	<code>r_shared</code> or <code>alpha_shared</code>
<code>eff_model</code>	Efficacy carryover model	<code>r_dose</code> , <code>alpha_dose</code> , <code>r_shared</code> , or <code>alpha_shared</code>
<code>tite</code>	Partial-outcome implementation	If TRUE, toxicity uses fixed- r_T TITE; efficacy uses delayed outcome
r_T^{fix}	Fixed toxicity carryover in TITE	User specified; $r_T^{\text{fix}} = 0$ gives original TITE-CRM
τ_T, τ_E	Endpoint assessment windows	User specified
<code>s1_Max</code>	Stage I sample-size limit	User specified
m_U	One-stage stay-rule threshold	User specified; current example $m_U = 6$
ϕ_T, c_T	Toxicity target and posterior cut-off	User specified
ϕ_E, c_E	Efficacy futility threshold and cutoff	User specified
δ	Minimum efficacy improvement for IPDE	User specified
<code>max_cycle</code>	Maximum cycles per patient	User specified
<code>flexible</code>	Permitted IPDE destination range	FALSE: upward only; TRUE: any admissible dose
<code>enrollment_scheme</code>	Cohort queue mechanism	Continuous new-patient priority or IPDE-first recruitment
<code>utility_type</code>	Dose-ranking function	1, 2, or 3
λ_T	Toxicity penalty in Utility 2	User specified
<code>tried</code>	Final-selection restriction	TRUE by default for MTD and OBD
Beta/Gamma shapes	Priors for $p_{E,j}$ and estimated carryover	Prespecified by model option; common Gamma rate for ratio representation

A. Implementation Templates

The following JAGS-style templates show the defining structure of the six carryover options. They are schematic and omit monitoring calculations and posterior-generated quantities.

A.1 Toxicity T-R: shared r_T

```
for (j in 1:ndose) {
  p_tox[j] <- pow(skeleton[j], exp(beta))
  q_tox_r[j] <- r_t * (1 - p_tox[j]) + p_tox[j]
}
for (i in 1:N) {
  theta_tox[i] <- (1 - ipde[i]) * p_tox[dose[i]] +
    ipde[i] * q_tox_r[dose[i]]
  tox[i] ~ dbern(theta_tox[i])
}
```

For the TITE implementation, supply r_t as fixed data and replace the last line with $\text{tox}[i] \sim \text{dbern}(w_{\text{tox}}[i] * \theta_{\text{tox}}[i])$, forcing $w_{\text{tox}}[i]=1$ for observed DLTs.

A.2 Toxicity T-A: shared α_T

```
g_tox_carry_1 ~ dgamma(a_alpha_t, 1)
g_tox_carry_2 ~ dgamma(b_alpha_t, 1)
alpha_t <- g_tox_carry_1 / (g_tox_carry_1 + g_tox_carry_2)

for (j in 1:ndose) {
  p_tox[j] <- pow(skeleton[j], exp(beta))
}
for (i in 1:N) {
  q_tox_alpha[i] <- min(1,
    p_tox[dose[i]] + alpha_t * p_tox[previous_dose[i]])
  theta_tox[i] <- (1 - ipde[i]) * p_tox[dose[i]] +
    ipde[i] * q_tox_alpha[i]
  tox[i] ~ dbern(theta_tox[i])
}
```

A.3 E-RD: dose-specific $r_{E,j}$

```
for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
    (g_regular_1[j] + g_regular_2[j])

  g_carry_1[j] ~ dgamma(a_carry[j], 1)
  g_carry_2[j] ~ dgamma(b_carry[j], 1)
  r_e[j] <- g_carry_1[j] / (g_carry_1[j] + g_carry_2[j])
  p_ipde[j] <- r_e[j] + (1 - r_e[j]) * p_regular[j]
}
for (i in 1:N) {
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
    ipde[i] * p_ipde[dose[i]]
}
```

A.4 E-AD: dose-specific $\alpha_{E,j}$

```
for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
    (g_regular_1[j] + g_regular_2[j])

  g_alpha_1[j] ~ dgamma(a_alpha[j], 1)
  g_alpha_2[j] ~ dgamma(b_alpha[j], 1)
  alpha_e[j] <- g_alpha_1[j] / (g_alpha_1[j] + g_alpha_2[j])
}
for (i in 1:N) {
  q_eff[i] <- min(1, p_regular[dose[i]] +
    alpha_e[dose[i]] * p_regular[previous_dose[i]])
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
    ipde[i] * q_eff[i]
}
```

A.5 E-RS: shared r_E

```

g_r_1 ~ dgamma(a_r, 1)
g_r_2 ~ dgamma(b_r, 1)
r_e <- g_r_1 / (g_r_1 + g_r_2)

for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
    (g_regular_1[j] + g_regular_2[j])
  p_ipde[j] <- r_e + (1 - r_e) * p_regular[j]
}
for (i in 1:N) {
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
    ipde[i] * p_ipde[dose[i]]
}

```

A.6 E-AS: shared α_E

```

g_alpha_1 ~ dgamma(a_alpha, 1)
g_alpha_2 ~ dgamma(b_alpha, 1)
alpha_e <- g_alpha_1 / (g_alpha_1 + g_alpha_2)

for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
    (g_regular_1[j] + g_regular_2[j])
}
for (i in 1:N) {
  q_eff[i] <- min(1, p_regular[dose[i]] +
    alpha_e * p_regular[previous_dose[i]])
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
    ipde[i] * q_eff[i]
}

```

A.7 Delayed-outcome efficacy likelihood in JAGS

A zeros-trick or equivalent custom likelihood is needed because the pending nonresponse contribution is $(1 - \theta_{E,i})^{w_{E,i}}$ rather than a standard Bernoulli term. Define

$$\ell_{E,i} = \Delta_{E,i} Y_{E,i} \log \theta_{E,i} + \{\Delta_{E,i}(1 - Y_{E,i}) + (1 - \Delta_{E,i})w_{E,i}\} \log(1 - \theta_{E,i}). \quad (38)$$

Then implement $-\ell_{E,i}$ through a zeros-trick with a sufficiently large constant. Numerical protection should bound $\theta_{E,i}$ away from exactly zero and one before taking logarithms.

B. References

- [1] Cheung, Y. K., and Chappell, R. (2000). Sequential designs for phase I clinical trials with late-onset toxicities. *Biometrics*, 56, 1177–1182.
- [2] Lin, R., and Yuan, Y. (2020). Time-to-event model-assisted designs for dose-finding trials with delayed toxicity. *Biostatistics*, 21(4), 807–824.
- [3] O’Quigley, J., Pepe, M., and Fisher, L. (1990). Continual reassessment method: a practical design for phase I clinical trials in cancer. *Biometrics*, 46, 33–48.
- [4] Updated TITE-CRM Likelihoods AIDE-CRM: Updated Models and Likelihood Construction. Internal technical note, July 12, 2026.
- [5] Current One-Stage and Two-Stage IPDE Design with Efficacy. Technical design specification, July–August 2026.

C. TITE Event-Clock Recycle Strategy

Scope. This section adopts the event-clock and waiting-queue strategy in the TITE-Likelihood technical note, with revised efficacy-enabled recycle eligibility. New patients are considered before eligible retreat patients. A patient becomes structurally eligible after completing the full DLT assessment without a DLT or efficacy response and may be recycled to any currently admissible dose, including the dose assigned to the regular cohort. No adjacency or one-level direction restriction is imposed. When no cohort is open, a new model-based decision remains unavailable until at least n_{eval} patients treated at the current dose have fully evaluated DLT outcomes.

C.1 Event-clock data structure

Calendar time is tracked explicitly. Each administration row records patient ID, start and evaluation times, assigned dose, cycle, cohort, and regular-versus-retreat type. At each decision attempt, toxicity and efficacy status and the applicable time-to-event weights are reconstructed from all administrations that have started.

C.2 Assignable events and waiting queue

An **assignable event** makes a treatment opportunity available for possible cohort filling:

1. a new-patient arrival; or
2. an IPDE/retreat decision at the end of a patient's assessment window. If the preceding administration completed the full DLT assessment without a DLT or efficacy response and the patient remains below the cycle limit, a retreat event is scheduled at that time.

If no cohort position is open, the event enters the waiting queue and is retained.

new	Newly arrived patient who has not yet received treatment.
retreat	Previously treated patient who completed the full DLT assessment without a DLT or efficacy response and may be available for IPDE.

The queue is ordered by event time, with sequence number breaking ties. When an open cohort position is filled, waiting new-patient events are considered first. Eligible retreat events are considered only after no waiting new patient remains. A retreat event that currently has no acceptable destination remains queued until an acceptable destination becomes available or the trial terminates.

C.3 Cohort formation and model-based decisions

The event loop is determined first by whether a cohort is already open.

Open cohort. If a cohort is open, no new regular-cohort dose decision is made. Available positions are filled from the waiting queue using new-patient-first priority. If fewer queued events are available than open positions, the cohort remains open and each later assignable event is used to fill the next position. The cohort closes only after **cohortsize** administration rows have been assigned.

No open cohort. If no cohort is open, every assignable event triggers a decision-availability check. Let $N_{eval}(d_{current})$ be the number of patients treated at the current dose whose DLT outcome is fully ascertained, either because a DLT has been observed or because the full DLT assessment window has been completed without DLT. The prespecified minimum is n_{eval} .

If $N_{eval}(d_{current}) < n_{eval}$, the decision is unavailable: no cohort is opened, no new regular-cohort dose is assigned, and all waiting events remain queued. Each later assignable event repeats this check. The starting cohort at the prespecified initial dose is opened according to the trial initialization rule and is not blocked by this gate.

Once $N_{eval}(d_{current}) \geq n_{eval}$, the interim toxicity and efficacy data are reconstructed, the applicable time-to-event weights are calculated, and the prespecified toxicity and efficacy models defined earlier in this document are updated. The active one-stage or two-stage design algorithm then applies its overtotoxicity, futility, admissibility, no-skipping, dose-cap, suspension, and stopping rules.

If the active algorithm does not return an assignable regular-cohort dose, no cohort is opened and all waiting events remain queued. If a regular-cohort dose $d_{current}$ is returned, a new cohort is opened and its available positions are filled immediately from the queue. A retreat patient used to fill a position is evaluated under Section C.4 and may receive $d_{current}$ or another admissible destination.

C. TITE Event-Clock Recycle Strategy (continued)

Operational interpretation. Open cohort $\rightarrow$ fill it. No open cohort $\rightarrow$ at each assignable event, first check whether at least n_{eval} patients have fully evaluated DLT outcomes at the current dose. Threshold not met $\rightarrow$ decision unavailable and retain the queue. Threshold met $\rightarrow$ apply the prespecified models and the corresponding one-stage or two-stage algorithm. Decision returned $\rightarrow$ open a new cohort and fill it.

C.4 Retreat/IPDE eligibility and destination set

A waiting retreat event is structurally eligible for reassignment when:

1. the patient has a previous administration record;
2. the most recent administration has completed the full DLT assessment window;
3. no DLT was observed during that completed administration;
4. no efficacy response was observed during that completed administration; and
5. completed cycles are below cycle_max .

Structural eligibility is checked when a retreat event is considered for an open cohort position. Let $\mathcal{A}$ be the current admissible dose set produced by the most recent model update and the active one-stage or two-stage algorithm. Every $k \in \mathcal{A}$ may be considered as a recycle destination, including a lower dose, the patient's preceding dose, a higher dose, and the regular-cohort dose d_{current} . There is no requirement that $k = d_{\text{current}}$ or $k = d_{\text{last}} + 1$.

For each candidate k , calculate the IPDE toxicity and efficacy probabilities under the prespecified toxicity and efficacy model options selected earlier in this document. The patient may be assigned to k only if the model-specific IPDE toxicity-risk and expected efficacy-improvement criteria are satisfied. If no candidate satisfies these criteria, the retreat event is not used for the current position and remains queued or is removed according to the prespecified queue rule.

C.5 Operational sequence

At each assignable event:

1. update the event clock and add the new-patient or completed-cycle retreat opportunity to the queue;
2. if a cohort is open, fill its available positions from the queue using new-patient-first priority; do not make a new regular-cohort dose decision;
3. if no cohort is open, count $N_{\text{eval}}(d_{\text{current}})$, the patients at the current dose with fully ascertained DLT outcomes;
4. if $N_{\text{eval}}(d_{\text{current}}) < n_{\text{eval}}$, declare the decision unavailable, keep the cohort closed, retain the waiting queue, and wait for the next assignable event;
5. if $N_{\text{eval}}(d_{\text{current}}) \geq n_{\text{eval}}$, reconstruct the interim data, calculate the applicable time-to-event weights, update the selected toxicity and efficacy models, and apply the active one-stage or two-stage algorithm;
6. if the algorithm returns no assignable dose, keep the cohort closed and retain the waiting queue;
7. if the algorithm returns d_{current} , open a new cohort and fill it from the queue; for each retreat event considered, evaluate all currently admissible destinations under Section C.4; and
8. continue until the administration cap or another stopping condition is reached.

C.6 Relationship to efficacy-based IPDE criteria

The event-clock rules determine *when* a patient becomes available and when a cohort decision is attempted. The selected toxicity and efficacy models determine the patient-specific IPDE risk and efficacy-improvement quantities. The n_{eval} gate first determines whether a model-based regular-cohort decision may be attempted. Once that gate is met, the active one-stage or two-stage algorithm determines whether an assignable regular-cohort dose is returned and defines the current admissible dose set. Queue priority determines only which waiting event is considered first; it does not determine the dose.

Implementation rule. Any currently admissible dose may be selected for an eligible retreat patient, including the regular-cohort dose. New-patient-first priority does not override destination-specific toxicity, futility, or efficacy-improvement criteria.

C.7 Queue behavior and edge cases

- If one or more new patients are waiting, the earliest waiting new patient is assigned before any retreat event.
- A retreat event with no currently acceptable destination is not automatically discarded; it may remain queued until an acceptable destination becomes available or the trial ends.
- If a cohort remains partially filled, later assignable events fill the open positions without a new dose decision.
- When no cohort is open and fewer than n_{eval} patients have fully evaluated DLT outcomes at the current dose, each assignable event leaves the decision unavailable and the queue unchanged except for the newly added event.
- The administration cap counts both regular and retreat administrations.
- A patient can generate another retreat event only after the latest administration completes the full DLT assessment without a DLT or efficacy response and the cycle limit has not been reached.

C.8 Required reporting

Report queue entries and waiting times by event type; the value of $N_{\text{eval}}(d_{\text{current}})$ at each closed-cohort assignable event; events at which the n_{eval} gate prevents a decision; model-based decision attempts with and without an assignable regular-cohort dose; retreat eligibility and reasons; preceding and destination doses; assignments to d_{current} versus other admissible doses; events queued at termination; and total regular plus retreat administrations.